

All cancers combined, there was 7% local recurrence in the treatment field out of 13% local recurrence. Sixty-two lesions had a complete response after treatment (49%). Thirty-nine percent of patients presented acute cutaneous toxicities (radioepithelitis) with 35% of grade I-II, 1.5% of grade III and 0.08% of grade IV.

Fifty-six lesions treated were skin cancers. Fifty percent of this patients presented toxicities with exclusively grade I-II toxicities. In Basal cell carcinoma (32 lesions), we have noted 47% of toxicities with exclusively grade I-II. In carcinoma epidermoid (15 lesions), we found 53% of toxicities with exclusively grade I-II.

Seventy lesions treated were skin metastasis. Thirty-one percent of patients presented toxicities with 24% of grade I-II, 3% of grade III and 1,4% grade IV toxicities (necrosis). In melanoma metastasis (23 lesions), we have noted 30% grade I-II, 4% grade III. In others metastasis (47 lesions), we found 21% grade I-II, 2% grade III and 2% grade IV.

Analyses showed no correlation between gender or tumor histology and the occurrence or grade of acute toxicities, with r correlation coefficients at 0.00 ($p=1$).

Conclusion

High dose rate brachytherapy is a technique that combines effectiveness with a good local control rate and good clinical tolerance. The results are comparable whether the treatment is for a skin cancer or a cutaneous metastasis.

PO-1810 Postoperative Management of Keloids with High-Dose-Rate Brachytherapy

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Purpose or Objective

The effectiveness of postoperative interventional radiotherapy (POIRT) in preventing keloid recurrences after surgical excision has already been confirmed. We report our updated monocentric experience on the adjuvant high-dose-rate (HDR) interstitial brachytherapy.

Materials and Methods

We retrospectively analyzed the medical records of 133 consecutive patients (104 female; 29 male) with 172 keloids. Patients receiving complete surgical excision and HDR-POIRT treatment from October 2004 to February 2021. Median age at the time of treatment was 43 years (range 16 - 76). During the surgical procedure, at the end of the scar excision, the dedicated plastic tube for brachytherapy was inserted through the center of the wound. Surgical wound was considered as the target volume, and the irradiation usually started within 4-6 hours after surgery. The scheduled dose was 12 Gy in 4 fractions, two fraction per day with at least 6 hours in-between. Recurrence was defined as the presence of a new keloid in a previously treated site (both along the scar or only at its extremity).

Results

Twenty-one patients were lost at follow-up, therefore the data of 112 patients and 147 keloids are presented. No complications, such as bleeding or scar infections, occurred either during treatments or after catheter removal. The most frequent early toxicity was erythema (18%), but most of the patients had no toxicity (80%). Late toxicity was registered in 22% of the 33 treated keloids. Over a median follow-up time of 40 months, 44 treated keloids (26%) relapsed. The median time of relapse was 8 months. Three patients received a re-treatment (surgery and brachytherapy) after the relapse of the keloid; after re-treatment no further relapses were registered.

Conclusion

Brachytherapy for keloids adjuvant treatment after surgical excision is an effective and safe treatment. Use of relatively low brachytherapy dose may partially explain the relapse rate observed in our series. Further prospective investigation with different treatment schedules will be planned in order to establish the optimal dose that should be employed in the postoperative treatment of this benign condition.

PO-1811 Efficacy of Episcleral Brachytherapy in Choroidal Melanoma

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Purpose or Objective

The purpose of this study is to evaluate the results of the treatment of Choroidal Melanoma with episcleral brachytherapy with I-125, in order to evaluate primarily the local control of the disease, metastasis-free survival, rates of enucleation and secondarily establish global survival and side effects of treatment.